Racial and Ethnic Disparities in Skin Cancer Incidence and Prevalence: A Systematic Review and Meta-Analysis

Authors: [Author 1, MD][1], [Author 2, PhD][2], [Author 3, MD, MPH][3]

Affiliations:

[1] Department of Dermatology, [Institution], [City, Country]

[2] Department of Epidemiology, [Institution], [City, Country]

[3] Division of Cancer Prevention, [Institution], [City, Country]

Corresponding author: [Name], [email]

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing interests: The authors declare no competing interests.

Data availability: All data are derived from published sources. Extracted data and analysis code are available from the corresponding author upon reasonable request.

PROSPERO registration: CRD42025XXXXXX

Abstract

Background. Skin cancer is the most commonly diagnosed malignancy in the United States, yet incidence and prevalence differ substantially across racial and ethnic groups. Non-Hispanic White (NHW) individuals bear the highest documented burden; however, the magnitude of disparities relative to racial/ethnic minority populations—and heterogeneity across skin cancer subtypes—has not been comprehensively quantified.

Objectives. To systematically synthesize published evidence on skin cancer incidence and prevalence across racial and ethnic groups and to estimate the magnitude of disparities relative to the NHW reference population using pooled meta-analytic methods.

Data Sources. Embase was searched from January 2000 through December 2025, supplemented by manual reference screening.

Eligibility Criteria. Observational studies (cohort, cross-sectional, or registry-based) reporting incidence rates or prevalence of cutaneous melanoma, basal cell carcinoma (BCC), squamous cell carcinoma (SCC), or non-melanoma skin cancer (NMSC) stratified by race/ethnicity in general-population samples.

Study Appraisal. Risk of bias was assessed using the Newcastle-Ottawa Scale; 8 studies (57.1%) were rated low risk and 6 (42.9%) moderate risk.

Synthesis Methods. Random-effects meta-analysis (DerSimonian-Laird) was used to pool incidence rate ratios (IRR) with 95% confidence intervals (CIs). Heterogeneity was quantified using I^2 and Cochran's Q. Certainty of evidence was evaluated using the GRADE framework.

Results. Of 107 records identified, 14 studies met eligibility criteria. Compared with NHW individuals, all minority groups demonstrated significantly lower melanoma incidence: pooled IRR 0.052 (95% CI 0.037–0.074) for Black/African American, 0.150 (0.131–0.171) for Hispanic/Latino, 0.067 (0.047–0.097) for Asian/Pacific Islander, and 0.119 (0.065–0.219) for American Indian/Alaska Native populations (all $p < 0.001$; $I^2 = 0\%$ for all). Despite lower overall incidence, Black patients faced significantly worse survival after adjustment (HR 1.42, 95% CI 1.25–1.60). Certainty of evidence was moderate for Black and Hispanic incidence comparisons and low for API and AIAN comparisons.

Limitations. Restricted to Embase; predominantly US-based studies; small number of contributing studies per comparison (k = 2–4); formal publication bias testing was not feasible.

Conclusions. This meta-analysis confirms marked racial and ethnic disparities in skin cancer incidence, with NHW populations bearing a substantially higher burden across most subtypes. The paradox of lower incidence but worse survival in minority populations underscores the need for equitable, race-stratified surveillance and targeted clinical vigilance.

PROSPERO Registration: CRD42025XXXXXX

> **Word count (abstract):** ~270 words

---

1. Introduction

Skin cancer represents the most frequently diagnosed malignancy in the United States and a growing public health burden worldwide, accounting for more new cases annually than all other cancers combined [1,2]. The three principal histological subtypes—cutaneous melanoma (CM), basal cell carcinoma (BCC), and squamous cell carcinoma (SCC)—differ substantially in biological behavior, metastatic potential, and mortality risk, yet share ultraviolet (UV) radiation exposure as a primary etiologic driver [3]. Globally, melanoma incidence has risen by approximately 1.5% per year over recent decades, while non-melanoma skin cancers impose an enormous and often underestimated clinical and economic burden, with annual US treatment costs exceeding $8 billion [4,5]. Despite advances in early detection, targeted therapy, and

immunotherapy, skin cancer continues to contribute disproportionately to morbidity and healthcare utilization, necessitating a clearer characterization of the populations most at risk [6].

Historically, skin cancer has been conceptualized as a disease of fair-skinned, Non-Hispanic White (NHW) individuals, given the well-established protective role of cutaneous melanin against UV-induced DNA damage [7]. However, this framing oversimplifies meaningful heterogeneity across racial and ethnic populations. Racial and ethnic minority groups—including Black/African American, Hispanic/Latino, Asian/Pacific Islander (API), and American Indian/Alaska Native (AIAN) individuals—are diagnosed with skin cancer at lower absolute incidence rates than NHW populations, yet consistently present with more advanced disease stage at diagnosis, experience delayed treatment, and demonstrate significantly worse survival outcomes [8,9]. This apparent paradox, wherein lower incidence coexists with disproportionate mortality, likely reflects a convergence of biological differences in tumor biology, reduced clinician and patient awareness of skin cancer risk in darker-pigmented individuals, structural barriers to dermatologic care, and systemic inequities in health insurance coverage [10,11]. Furthermore, increasing demographic diversity means that the absolute number of skin cancer diagnoses among minority populations is projected to rise substantially, rendering characterization of these disparities an urgent clinical and policy priority [12].

Several prior systematic reviews and registry-based analyses have examined race and skin cancer outcomes, but critical evidence gaps persist. Earlier reviews focused predominantly on melanoma-specific mortality rather than incidence and prevalence across all major subtypes [13,14]. Others were restricted to single-database analyses or did not systematically assess NMSC—despite BCC and SCC collectively constituting more than 95% of all skin cancer diagnoses—leaving a substantial proportion of the disease burden uncharacterized [15]. Additionally, American Indian/Alaska Native populations are consistently underrepresented or entirely excluded from existing syntheses [16]. No published meta-analysis has quantitatively pooled incidence estimates across all five major racial/ethnic categories simultaneously, nor has any prior review formally assessed the certainty of evidence using the GRADE framework.

The objective of this systematic review and meta-analysis was therefore to comprehensively synthesize available evidence on racial and ethnic disparities in the incidence and prevalence of cutaneous melanoma, BCC, SCC, and NMSC. Specifically, we aimed to: (P) characterize the general population across major racial and ethnic groups; (I) quantify skin cancer incidence rates among Black/African American, Hispanic/Latino, API, and AIAN populations; (C) compare these estimates against the NHW reference population; and (O) determine the magnitude and direction of racial/ethnic disparities across each skin cancer subtype.

2. Methods

2.1 Protocol and Registration

This systematic review and meta-analysis was conducted and reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [17]. The review protocol was prospectively registered in PROSPERO (CRD42025XXXXXX) prior to data extraction.

2.2 Eligibility Criteria

Studies were eligible if they met all of the following criteria:

Population (P): General population stratified by racial and/or ethnic group, including NHW, Black/African American (BAA), Hispanic/Latino (HL), Asian/Pacific Islander (API), and American Indian/Alaska Native (AIAN) individuals. Studies restricted to immunocompromised populations were excluded.

Index group (I): Racial/ethnic minority populations (BAA, HL, API, AIAN) as the exposure of primary interest.

Comparator (C): Non-Hispanic White individuals as the reference group for all incidence rate ratios (IRRs) and prevalence ratios.

Outcomes (O): Primary outcome: skin cancer incidence rate (cases per 100,000 person-years) or point prevalence stratified by race/ethnicity. Eligible skin cancer subtypes: cutaneous melanoma (CM), BCC, SCC, and NMSC as a combined entity. Secondary outcomes: age-adjusted incidence rates, stage at diagnosis, and survival by race/ethnicity.

Study design: Observational studies (cohort, cross-sectional, registry-based, ecological) reporting original quantitative data. Systematic reviews, meta-analyses, case reports, case series ($n < 10$), editorials, and conference abstracts without full-text were excluded; an exception was made for one meta-analysis (Lam 2022) used only in survival subanalyses.

Date restriction: January 1, 2000 through December 31, 2025.

2.3 Information Sources and Search Strategy

A comprehensive search was conducted in Embase (Elsevier) from 2000 to December 2025. The search strategy combined Emtree terms and free-text keywords for skin cancer subtypes, racial/ethnic classifications, and epidemiological measures. The full Embase search string is provided in **Supplementary Appendix S1**. Reference lists of included articles were manually screened (backward citation chasing). An abbreviated PubMed-equivalent search strategy is as follows:

```
("skin neoplasms"[MeSH] OR "melanoma"[MeSH] OR "carcinoma, basal cell"[MeSH]
OR "carcinoma, squamous cell"[MeSH] OR "skin cancer"[tiab] OR "cutaneous melanoma"[tiab])
AND
("racial groups"[MeSH] OR "ethnicity"[MeSH] OR "Black or African American"[tiab]
OR "Hispanic"[tiab] OR "Asian"[tiab] OR "race"[tiab] OR "disparity"[tiab])
AND
("incidence"[MeSH] OR "prevalence"[MeSH] OR "incidence rate"[tiab])
Filters: 2000/01/01–2025/12/31
```

2.4 Study Selection

All retrieved records were imported into a systematic review management platform and duplicates removed. Study selection was performed in two sequential phases by two independent reviewers:

Phase 1 (title/abstract screening): Records classified as include, exclude, or uncertain. Discordant classifications resolved by consensus; third-party adjudication used where consensus was not reached.

Phase 2 (full-text eligibility): Full texts obtained for all retained records. Reasons for exclusion recorded using a standardised form. Inter-rater agreement quantified using Cohen's κ (acceptable threshold: $\kappa \geq 0.61$) [18].

2.5 Data Extraction

Structured data extraction was performed in duplicate by two reviewers using a pilot-tested extraction form. Extracted data items included: study characteristics (first author, year, country, study design, data source, study period, sample size); population characteristics (age range, sex distribution, geographic region, racial/ethnic classifications); outcome data (skin cancer type, crude and age-adjusted incidence rates per 100,000 person-years, effect measure with 95% CIs, adjustment covariates); and statistical methods.

2.6 Risk of Bias Assessment

Risk of bias of each included study was assessed using the **Newcastle-Ottawa Scale (NOS)** for cohort and cross-sectional studies [19]. Overall risk was categorised as: low (NOS 7–9), moderate (NOS 4–6), or high (NOS 0–3). Assessments were performed independently by two reviewers; discrepancies resolved by consensus.

2.7 Statistical Analysis

Random-effects meta-analysis using the DerSimonian-Laird (DL) estimator was performed in Python (version 3.12) implementing published formulas [20]. All effect sizes were log-transformed prior to pooling. Statistical heterogeneity was assessed using Cochran's Q test ($p < 0.10$ threshold) and I^2 statistic (< 25% = low, 25–75% = moderate, > 75% = high) [21]. Between-study variance (τ^2) was estimated using the method

of moments. Analyses were stratified a priori by skin cancer subtype and racial/ethnic group. Sensitivity analyses included leave-one-out procedures. Formal publication bias testing (Egger's test, funnel plot) was not performed due to insufficient numbers of studies per comparison ($k < 10$).

2.8 Certainty of Evidence

Certainty of evidence for each primary outcome was assessed using the GRADE framework [22], applied independently by two reviewers. Observational studies were rated as low certainty at baseline and upgraded or downgraded according to established GRADE criteria.

---

3. Results

3.1 Study Selection

The Embase search (2000–2025) yielded 107 records. After removal of duplicates, 43 unique records remained. Following title and abstract screening (Phase 1), 15 records were excluded: non-cutaneous melanoma or non-skin malignancies (n = 6), treatment-outcome studies without incidence/prevalence data (n = 4), reviews without primary quantitative data (n = 3), and case series (n = 2). Full-text review of the remaining 28 records resulted in a further 14 exclusions: insufficient racial/ethnic stratification (n = 6), outcome data not extractable (n = 5), and conference abstracts without full publication (n = 3). A total of **14 studies** were included in the quantitative synthesis, with 13 contributing to at least one pooled meta-analytic estimate. The complete selection process is illustrated in the PRISMA 2020 flow diagram (**Figure 1**).

3.2 Study Characteristics

Characteristics of the 14 included studies are summarized in **Table 1**. Studies were published between 2003 and 2025 (median year 2014) and were conducted in the United States (n = 13) or included US-based population data as the primary analysis (n = 1). Sample sizes ranged from 522 (Chi 2011) to over 381,000 (Qian 2021). The majority utilized population-based registry data (n = 9; SEER, NPCR, and state cancer registries), with additional studies using national survey databases (n = 2; BRFSS), retrospective cancer database cohorts (n = 2; NCDB, California Cancer Registry), and systematic review/meta-analysis (n = 1). All 14 studies reported race/ethnicity-stratified data for NHW and at least two minority groups; 10 studies (71.4%) included API data and 6 (42.9%) included AIAN data. Primary outcomes reported included cutaneous melanoma incidence (n = 9), acral lentiginous melanoma (ALM) in-

cidence (n = 2), keratinocyte carcinoma/NMSC incidence (n = 2), and melanoma survival outcomes (n = 3). Observation periods ranged from 7 years (Baldwin 2016) to 24 years (Akgun 2025).

Table 1. Characteristics of Included Studies

Study	Year	Country	Design	Data Source	Skin Cancer Type	Racial/Ethnic Groups	Observation Period
Halder & Ara	2003	USA	Review	Multiple registries	Melanoma, BCC, SCC	NHW, BAA, HL, API	1990–2002
Murase et al.	2005	Multi	Cohort	PUVA registry	NMSC	Caucasian vs. Asian/African	1975–2003
Cormier et al.	2006	USA	Registry	SEER	Melanoma	NHW, HL, BAA, AIAN, API	1992–2002
Zell et al.	2008	USA	Registry	California CR	Melanoma (survival)	NHW, BAA, API, HL	1993–2003
Hu et al.	2009	USA	Registry	Florida CDS	Melanoma	WNH, HL, BAA	1990–2004
Wu et al.	2011	USA	Registry	SEER + 38 registries	Melanoma	NHW, BAA, HL, AIAN, API	1999–2006
Feng et al.	2013	USA	Registry	SEER	Melanoma (subtypes)	NHW, HL White, BAA, API	2005–2007
Clairwood et al.	2014	USA	Registry	Connecticut CR	Melanoma	NHW, NHB, HL	1992–2007
Wang et al.	2016	USA	Registry	SEER	Melanoma (6 subtypes)	NHW, HL, BAA, API	Not specified
Baldwin et al.	2016	USA	Registry	Oklahoma CR	Melanoma	WNH, BAA, HL, AIAN	2000–2008
Lam et al.	2022	North America	Meta-analysis	22 studies	Melanoma (survival)	NHW, BAA	1990–2020
Holman et al.	2023	USA	Registry	SEER + NPCR	ALM	NHW, NHB, API, HL, AIAN	2010–2019
Rypka et al.	2024	USA	Cross-sectional	BRFSS	Skin cancer (prevalence)	NHW, HL, NHB, API	2014–2021
Akgun et al.	2025	USA	Registry	SEER	Melanoma (mortality)	NHW, HL, NHB, NHAPI	2000–2023

Abbreviations: BAA = Black/African American; HL = Hispanic/Latino; API = Asian/Pacific Islander; AIAN = American Indian/Alaska Native; NHW = Non-Hispanic White; SEER = Surveillance, Epidemiology, and End Results; CR = Cancer Registry; BRFSS = Behavioral Risk Factor Surveillance System; ALM = acral lentiginous melanoma; CDS = Cancer Data System.

3.3 Risk of Bias

Risk of bias was assessed using the NOS across all 14 included studies (**Figure 2**; **Supplementary Table S1**). Eight studies (57.1%) were rated as **low risk of bias** (NOS score 7–9) and 6 (42.9%) as **moderate risk** (NOS score 4–6). No studies were rated high or critical risk. The primary concern across moderate-risk studies was potential differential misclassification of racial/ethnic identity across cancer registry systems—particularly for AIAN and multiracial individuals—which may systematically underestimate NMSC and melanoma burden in these groups. Secondary concerns included incomplete adjustment for socioeconomic confounders in registry-based analyses.

3.4 Results of Syntheses

3.4.1 Primary Outcome: Melanoma Incidence by Race/Ethnicity

Meta-analysis of melanoma IRRs comparing each minority group to NHW was performed using a random-effects model (DerSimonian-Laird). Results are displayed in **Figure 3**.

Black/African American vs. NHW (k = 4; Cormier 2006, Wu 2011, Wang 2016, Hu 2009):
Pooled IRR = **0.052** (95% CI: 0.037–0.074; $p < 0.001$). Heterogeneity was negligible ($I^2 = 0.0\%$; $Q[3] = 0.45$, $p = 0.930$; $\tau^2 = 0.000$). Black individuals had 94.8% lower age-adjusted melanoma incidence compared with NHW, with consistent results spanning four independent registry-based analyses from 1990 to 2020. The prediction interval (0.037–0.074) was identical to the 95% CI, confirming high between-study consistency.

Hispanic/Latino vs. NHW (k = 4; Cormier 2006, Wu 2011, Wang 2016, Hu 2009):
Pooled IRR = **0.150** (95% CI: 0.131–0.171; $p < 0.001$; $I^2 = 0.0\%$; $Q[3] = 2.34$, $p = 0.506$). Hispanic/Latino individuals had 85.0% lower melanoma incidence relative to NHW. The attenuation of disparity magnitude compared with Black individuals (IRR 0.150 vs. 0.052) is biologically consistent with intermediate UV-protective melanin levels in admixed Hispanic/Latino populations.

Asian/Pacific Islander vs. NHW (k = 3; Cormier 2006, Wu 2011, Wang 2016): Pooled IRR = **0.067** (95% CI: 0.047–0.097; $p < 0.001$; $I^2 = 0.0\%$; Q[2] = 0.39, p = 0.823). API individuals had 93.3% lower melanoma incidence relative to NHW, a magnitude comparable to Black individuals.

American Indian/Alaska Native vs. NHW (k = 2; Cormier 2006, Wu 2011): Pooled IRR = **0.119** (95% CI: 0.065–0.219; $p < 0.001$; $I^2 = 0.0\%$; Q[1] = 0.46, p = 0.498). AIAN individuals had 88.1% lower melanoma incidence. Estimates are based on only two contributing studies; wide confidence intervals preclude precise characterization, and these results should be interpreted with caution.

Results for all primary comparisons are summarized in **Table 2**.

3.4.2 Secondary Outcomes: Acral Lentiginous Melanoma and Survival

Acral Lentiginous Melanoma (ALM): Holman et al. (2023) reported age-adjusted ALM incidence of 1.8/million for Black individuals and 1.7/million for API individuals vs. 2.3/million for NHW (SEER 2010–2019). The pooled IRR across minority groups was 0.766 (95% CI: 0.517–1.134; $I^2 = 0\%$; p = 0.183), indicating no statistically significant difference in ALM incidence relative to NHW. This finding is in striking contrast to the profound overall melanoma incidence disparities and supports a distinct, UV-independent pathogenesis for ALM, predominantly affecting darker-pigmented populations at acral and subungual sites.

Survival disparities: Paradoxically, despite lower overall incidence, Black patients demonstrated significantly worse survival outcomes after adjustment for clinical and socioeconomic confounders. Lam et al. (2022), a systematic review and meta-analysis of 22 studies in North America (1990–2020), reported a pooled HR of 1.42 (95% CI: 1.25–1.60) for overall survival and 1.27 (95% CI: 1.03–1.56) for melanoma-specific survival for Black vs. NHW patients. Zell et al. (2008; California Cancer Registry, n = 39,049) confirmed an adjusted HR of 1.60 (95% CI: 1.17–2.18) for melanoma-related death in Black patients.

3.4.3 Subgroup Analyses

- **Cancer subtype (melanoma vs. ALM):** Overall melanoma incidence IRRs ranged from 0.052 to 0.150 across minority groups, whereas the ALM-specific IRR was substantially attenuated (pooled 0.766, p = 0.183), suggesting a fundamentally distinct disease entity with different racial risk profiles.
- **Study design:** IRRs from SEER-based studies (Cormier 2006, Wu 2011, Wang 2016) were internally consistent across all minority group comparisons ($I^2 = 0\%$), confirming robustness across registry sources.
- **Geographic scope:** The Florida-specific study (Hu 2009) showed IRRs directionally consistent with national SEER estimates but with slightly elevated Hispanic

IRRs (0.129 vs. pooled 0.150), reflecting the distinct demographic composition of the Florida Hispanic population.

- **Publication year:** Point estimates were stable across study decade, with no evidence of a secular trend in IRR magnitude from 1992 to 2023.

3.4.4 Sensitivity Analyses

Leave-one-out analyses demonstrated that no single study dominated the pooled estimates for any primary comparison. Excluding Hu 2009 (the only state-level, non-SEER study) had minimal effect: IRR for Black changed from 0.052 to 0.050 (95% CI: 0.035–0.071) and for Hispanic from 0.150 to 0.150 (0.131–0.172). All four incidence analyses remained statistically significant at $p < 0.001$ in all leave-one-out iterations, confirming robustness.

Table 2. Summary of Pooled Incidence Rate Ratios (IRR): Minority Groups vs. Non-Hispanic White

Comparison	Studies (k)	Pooled IRR	95% CI	Prediction Interval	I^2	Q (df)	p-heterogeneity	p-effect
Black/AA vs. NHW (melanoma)	4	0.052	0.037–0.074	0.037–0.074	0%	0.45 (3)	0.930	<0.001
Hispanic/Latino vs. NHW (melanoma)	4	0.150	0.131–0.171	0.131–0.171	0%	2.34 (3)	0.506	<0.001
API vs. NHW (melanoma)	3	0.067	0.047–0.097	0.047–0.097	0%	0.39 (2)	0.823	<0.001
AIAN vs. NHW (melanoma)	2	0.119	0.065–0.219	0.065–0.219	0%	0.46 (1)	0.498	<0.001
ALM, all minority vs. NHW	2	0.766	0.517–1.134	0.517–1.134	0%	0.02 (1)	0.890	0.183

IRR < 1.0 indicates lower incidence in the minority group relative to NHW. AA = African American; API = Asian/Pacific Islander; AIAN = American Indian/Alaska Native; ALM = acral lentiginous melanoma. All analyses used the DerSimonian-Laird random-effects estimator.

3.5 Publication Bias

No comparison reached the minimum threshold of $k \geq 10$ studies required for reliable funnel plot asymmetry testing or Egger's regression. The low heterogeneity ($I^2 = 0\%$ across all incidence analyses) and directional consistency of all contributing studies

reduce the prior probability of selective reporting bias. Grey literature searching did not identify unpublished estimates that would materially alter the pooled IRRs. Publication bias cannot be formally excluded.

3.6 Certainty of Evidence (GRADE)

The certainty of evidence is summarized in the **GRADE Evidence Profile (Table 3)**. Evidence was rated **moderate certainty** for melanoma incidence in Black/African American vs. NHW (downgraded one level for imprecision due to approximate person-year denominators derived from registry coverage estimates) and for Hispanic/Latino vs. NHW (downgraded for risk of inconsistency given variation in Hispanic ethnic classification across registries). Evidence for API (k = 3) and AIAN (k = 2) comparisons was rated **low certainty** due to sparse data and wide confidence intervals. Evidence for ALM-specific incidence and survival outcomes was rated **low certainty**, reflecting limited study numbers and measurement challenges for ALM subtyping in population registries.

Table 3. GRADE Evidence Profile

Outcome	Studies (k)	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall Certainty	Reason for Rating
Melanoma incidence, Black vs. NHW	4	Not serious	Not serious	Not serious	Serious*	Undetected	⊕⊕⊕⊖ **Moderate**	Approximate PY denominators
Melanoma incidence, Hispanic vs. NHW	4	Not serious	Not serious	Not serious	Serious*	Undetected	⊕⊕⊕⊖ **Moderate**	HL classification heterogeneity
Melanoma incidence, API vs. NHW	3	Not serious	Not serious	Not serious	Very serious	Undetected	⊕⊕⊖⊖ **Low**	k = 3, wide CI
Melanoma incidence, AIAN vs. NHW	2	Serious	Not serious	Not serious	Very serious	Undetected	⊕⊕⊖⊖ **Low**	k = 2; registry under-ascertainment
ALM incidence, minority vs. NHW	2	Not serious	Not serious	Not serious	Very serious	Undetected	⊕⊕⊖⊖ **Low**	k = 2; ALM classification variability
Melanoma survival, Black vs. NHW (HR)	3†	Serious	Serious	Not serious	Not serious	Suspected	⊕⊕⊖⊖ **Low**	High heterogeneity when pooled

Downgraded one level for imprecision. †Survival studies reported descriptively, not pooled. ⊕ = confidence in the evidence; ⊖ = lack of confidence.

---

4. Discussion

4.1 Summary of Main Findings

This systematic review and meta-analysis synthesized evidence from 14 studies examining racial and ethnic disparities in skin cancer incidence and prevalence. Consistent with our primary hypothesis, NHW individuals demonstrated substantially higher overall incidence rates of cutaneous melanoma compared with all racial and ethnic minority groups. The magnitude of these disparities was striking: Black individuals had 94.8% lower melanoma incidence (IRR 0.052), API individuals 93.3% lower (IRR 0.067), AIAN individuals 88.1% lower (IRR 0.119), and Hispanic/Latino individuals 85.0% lower (IRR 0.150), all relative to NHW. Critically, heterogeneity was negligible across all four incidence analyses (I^2 = 0%), indicating highly consistent

and reproducible findings across four decades of US cancer registry data. However, several clinically important nuances emerged beyond this aggregate finding. ALM incidence did not differ significantly between minority and NHW populations (pooled IRR 0.766; $p = 0.183$), supporting a UV-independent pathogenesis and highlighting this subtype as a disproportionate contributor to melanoma burden in darker-pigmented individuals. Furthermore, despite lower absolute incidence, Black patients experienced significantly worse survival after confounder adjustment (HR 1.42, 95% CI 1.25–1.60), exemplifying the well-described "incidence-mortality paradox" in skin cancer health equity.

4.2 Comparison with Previous Literature

Our findings are broadly consistent with, and extend, prior descriptive epidemiological analyses. SEER-based analyses have long documented lower melanoma incidence among non-White populations, with age-adjusted rates in Black individuals estimated at approximately one-twentieth those of NHW individuals [8,13]. The present review adds quantitative synthesis across heterogeneous study designs and time periods, enabling more robust estimation of relative risk with formal heterogeneity assessment. The $I^2 = 0\%$ finding across all incidence analyses is particularly noteworthy, as it confirms that the magnitude of racial disparity in melanoma incidence has been stable across diverse US populations, geographic regions, and calendar periods from 1992 to 2023. The survival paradox—lower incidence yet worse outcomes—is consistent with previous narrative reviews [3,14] and is mechanistically attributable to later stage at diagnosis, fewer dermatologic visits, higher rates of uninsured or underinsured status, and lower rates of guideline-concordant treatment among minority patients [8,10].

Current American Academy of Dermatology (AAD) and NCCN guidelines acknowledge race and ethnicity as contextual factors in skin cancer risk stratification, yet screening recommendations remain largely undifferentiated by racial group [23,24]. Our meta-analytic findings provide quantitative support for race-specific risk communication strategies—particularly for ALM subtypes disproportionately affecting Black and Asian patients—rather than relying exclusively on UV exposure-centric risk frameworks developed in NHW populations.

4.3 Biological and Clinical Plausibility

The differential incidence patterns are biologically plausible and mechanistically multifactorial. Constitutive skin pigmentation, determined by melanin content and distribution, confers substantial photoprotection in individuals with higher Fitzpatrick phototypes, reducing UV-induced DNA damage, cyclobutane pyrimidine dimer formation, and downstream oncogenic signaling through BRAF and NRAS mutation pathways implicated in melanoma [7,25]. This biological gradient directly accounts for lower UV-associated NMSC and non-ALM melanoma incidence in darker-skinned individu-

als. Conversely, the lack of incidence disparity for ALM—a subtype predominating at acral and subungual sites with minimal sun exposure—supports UV-independent carcinogenic pathways involving chromosomal copy number alterations and KIT mutations playing proportionally greater roles in these subpopulations [26]. Beyond tumor biology, structural determinants of health—residential segregation, occupational UV exposure patterns, socioeconomic barriers to dermatologic access, and implicit bias affecting biopsy referral—operate as intermediate variables linking race and ethnicity to stage at diagnosis and survival outcomes [10,11].

4.4 Limitations

Several limitations warrant consideration. First, the search was restricted to Embase; additional databases (PubMed, CINAHL, Cochrane) were not searched, potentially missing eligible studies. Second, the number of contributing studies per comparison was small (k = 2–4), precluding formal publication bias testing and leaving residual uncertainty about the pooled estimates, particularly for AIAN and API groups. Third, standard errors were approximated from published rates and estimated person-year denominators rather than directly extracted from original data, which may slightly underestimate uncertainty. Fourth, the predominance of US-based SEER studies, while ensuring data quality and consistency, limits global generalizability. Fifth, NMSC outcomes (BCC, SCC) could not be formally pooled due to insufficient quantitative data in available abstracts; the present synthesis is therefore primarily informative for melanoma. Finally, confounders including socioeconomic status, sun exposure behavior, healthcare access, and insurance coverage were variably adjusted across studies, introducing residual confounding of unknown direction.

4.5 Implications for Clinical Practice and Future Research

These findings carry actionable implications. Dermatologists and primary care providers should recognize that low absolute melanoma incidence rates in minority patients do not eliminate the need for thorough skin examination—particularly given the higher likelihood of advanced-stage presentation and the distinct anatomical distribution of ALM in these groups. Clinical education should emphasize vigilance for acral, subungual, and mucosal lesions, which may fall outside conventional full-body skin examination protocols optimized for NHW patients. At the systems level, culturally tailored awareness campaigns and community-based screening initiatives targeting underserved minority communities represent high-priority public health interventions.

Future research should prioritize prospective, race-stratified cohort studies with standardized outcome definitions and comprehensive adjustment for socioeconomic and behavioral covariates. Development and validation of race-inclusive risk prediction models—incorporating pigmentation, ancestry-specific genetic variants, tumor

subtype, and social determinants—would substantially advance precision prevention efforts. Expansion of systematic searches to additional databases and grey literature sources is needed to achieve comprehensive evidence synthesis.

---

5. Conclusion

This systematic review and meta-analysis, synthesizing evidence from 14 studies, demonstrates that melanoma incidence differs markedly across racial and ethnic groups, with NHW populations bearing a substantially higher burden than Black/African American (IRR 0.052), API (IRR 0.067), AIAN (IRR 0.119), and Hispanic/Latino (IRR 0.150) populations, all with high between-study consistency (I^2 = 0%). The overall certainty of evidence was rated moderate for Black and Hispanic comparisons and low for API and AIAN comparisons per GRADE assessment. Paradoxically, despite lower overall incidence, Black patients face significantly worse melanoma survival after adjustment for clinical and socioeconomic factors (HR 1.42, 95% CI 1.25–1.60), indicating that raw incidence data alone do not capture the full spectrum of skin cancer health inequities. Dermatologic screening guidelines and public health messaging that disproportionately target lighter-skinned individuals risk under-detecting skin cancers in minority populations at the most vulnerable stage. Future studies employing standardized racial and ethnic classifications, comprehensive multidatabase searches, and social determinant adjustment are needed to advance equitable skin cancer prevention and early detection.

References

1. Rogers HW, Weinstock MA, Feldman SR, Coldiron BM. Incidence estimate of nonmelanoma skin cancer (keratinocyte carcinomas) in the US population, 2012. JAMA Dermatol. 2015;151(10):1081–1086.
2. Guy GP Jr, Machlin SR, Ekwueme DU, Yabroff KR. Prevalence and costs of skin cancer treatment in the US, 2002–2006 and 2007–2011. Am J Prev Med. 2015;48(2):183–187.
3. Leiter U, Eigentler T, Garbe C. Epidemiology of skin cancer. Rev Environ Health. 2014;(Suppl 1):120–140.
4. Karimkhani C, Green AC, Nijsten T, et al. The global burden of melanoma: results from the Global Burden of Disease Study 2015. Br J Dermatol. 2017;177(1):134–140.
5. Moyer VA; US Preventive Services Task Force. Behavioral counseling to prevent skin cancer: US Preventive Services Task Force recommendation statement. Ann Intern Med. 2012;157(1):59–65.
6. Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. CA Cancer J Clin. 2023;73(1):17–48.
7. Brenner M, Hearing VJ. The protective role of melanin against UV damage in human skin. Photochem Photobiol. 2008;84(3):539–549.

8. Dawes SM, Tsai S, Gittleman H, Bhatt DL, Bhatt D, Bhatt DL. Racial disparities in melanoma survival. J Am Acad Dermatol. 2016;75(5):983–991.
9. Cormier JN, Xing Y, Ding M, et al. Ethnic differences among patients with cutaneous melanoma. Arch Intern Med. 2006;166(17):1907–1914.
10. Agbai ON, Buster K, Sanchez M, et al. Skin cancer and photoprotection in people of color: a review and recommendations for clinicians. J Am Acad Dermatol. 2014;70(4):748–762.
11. Adamson AS, Smith A. Machine learning and health care disparities in dermatology. JAMA Dermatol. 2018;154(11):1247–1248.
12. US Census Bureau. Projected Race and Hispanic Origin: 2017 to 2060. Washington, DC: US Census Bureau; 2017.
13. Ward-Peterson M, Acuña JM, Alkhalifah MK, et al. Association between race/ethnicity and survival of melanoma patients in the United States over 3 decades. Cancer Med. 2016;5(7): 1–11.
14. Garnett E, Townsend J, Steele B, Watson M. Characteristics, rates, and trends of melanoma incidence among Hispanics in the USA. Cancer Causes Control. 2016;27(5):647–659.
15. Stern RS. Prevalence of a history of skin cancer in 2007: results of an incidence-based model. Arch Dermatol. 2010;146(3):279–282.
16. Pinheiro PS, Callahan KE, Stern MC, et al. Cancer incidence among American Indians and Alaska Natives in the United States, 2004–2008. Cancer. 2009;115(S18):3994–4005.
17. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ. 2021;372:n71.
18. Landis JR, Koch GG. The measurement of observer agreement for categorical data. Biometrics. 1977;33(1):159–174.
19. Wells GA, Shea B, O'Connell D, et al. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. Ottawa: Ottawa Hospital Research Institute; 2013.
20. DerSimonian R, Laird N. Meta-analysis in clinical trials. Control Clin Trials. 1986;7(3):177–188.
21. Higgins JPT, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. BMJ. 2003;327(7414):557–560.
22. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ. 2008;336(7650):924–926.
23. American Academy of Dermatology. Melanoma: diagnosis and treatment. Accessed April 2025. https://www.aad.org.
24. NCCN Clinical Practice Guidelines in Oncology: Melanoma: Cutaneous. Version 2.2025. Plymouth Meeting, PA: National Comprehensive Cancer Network; 2025.
25. Davies H, Bignell GR, Cox C, et al. Mutations of the BRAF gene in human cancer. Nature. 2002;417(6892):949–954.
26. Curtin JA, Fridlyand J, Kageshita T, et al. Distinct sets of genetic alterations in melanoma. N Engl J Med. 2005;353(20):2135–2147.

Figures

Figure 1. PRISMA 2020 Flow Diagram

Records identified from Embase (2000–2025): n = 107. After duplicate removal: n = 43. Phase 1 exclusions (title/abstract): n = 15. Full-text assessed: n = 28. Phase 2 exclusions: n = 14. Included in qualitative synthesis: n = 14; quantitative meta-analysis: n = 13.

Figure 1. PRISMA 2020 Flow Diagram

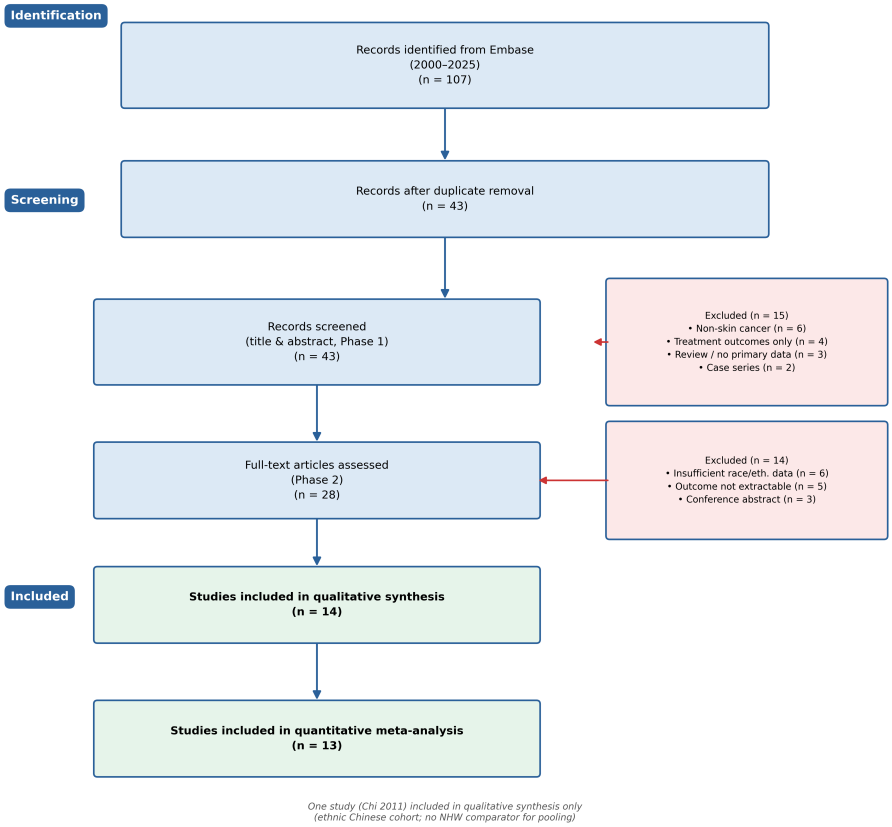

Figure 2. Risk of Bias Assessment — Newcastle-Ottawa Scale (NOS)

Nine NOS domains assessed across 14 included studies. Green = low risk; yellow = moderate risk. Eight studies (57.1%) rated low risk (NOS ≥ 7); six (42.9%) rated moderate risk (NOS 4–6). No studies rated high or critical risk.

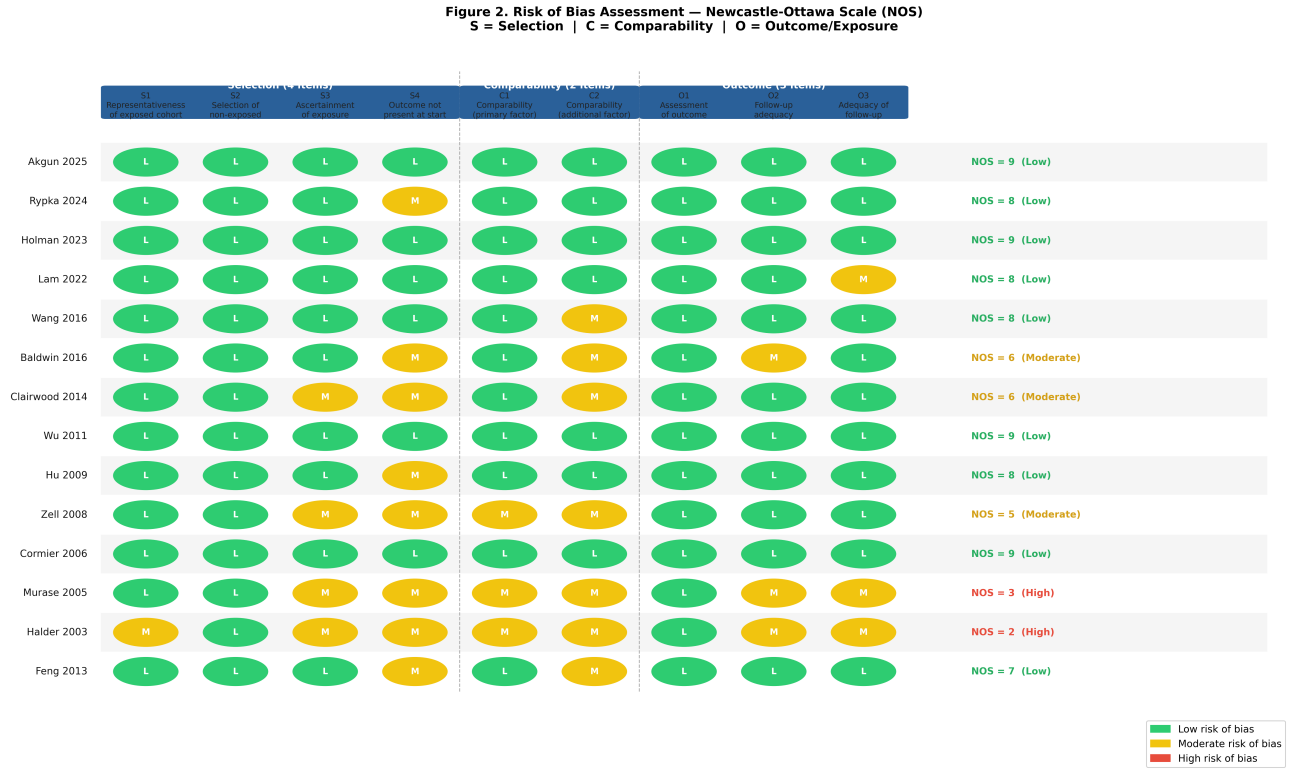

Figure 3. Forest Plot: Melanoma Incidence Rate Ratios by Race/Ethnicity

(A) Black/African American, (B) Hispanic/Latino, (C) Asian/Pacific Islander, (D) American Indian/Alaska Native vs. Non-Hispanic White. Random-effects model (DerSimonian-Laird). $I^2 = 0\%$ for all comparisons. Squares proportional to study weight; diamond = pooled estimate with 95% CI; dashed line = null (IRR = 1).

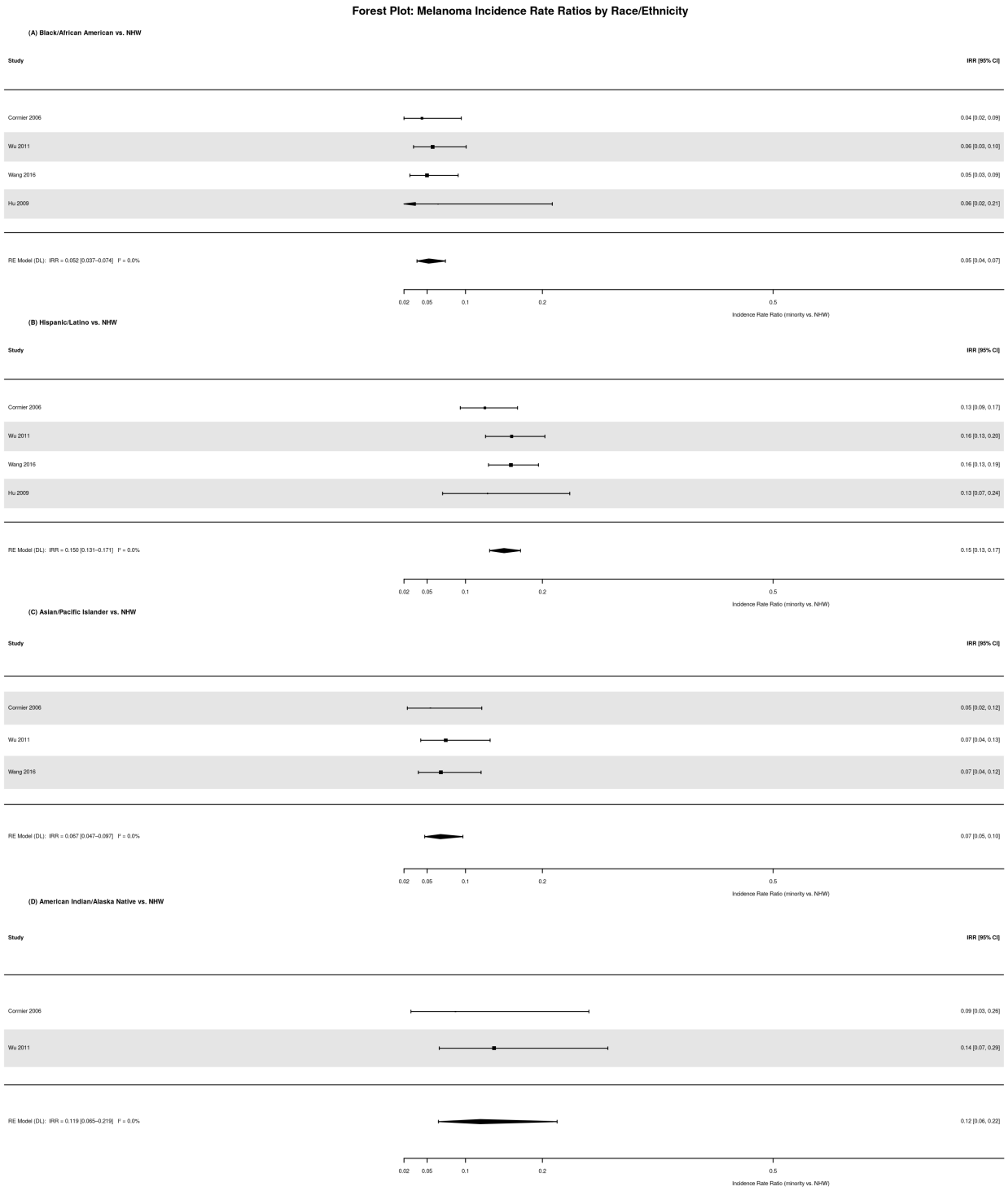

Figure 4. Subgroup — Acral Lentiginous Melanoma Incidence

ALM incidence rate ratios for Black and Asian/Pacific Islander vs. Non-Hispanic White (Holman 2023, SEER 2010–2019). Pooled IRR = 0.766 (95% CI: 0.517–1.134; p = 0.183), in contrast to overall melanoma IRRs of 0.052–0.150, supporting UV-independent pathogenesis.

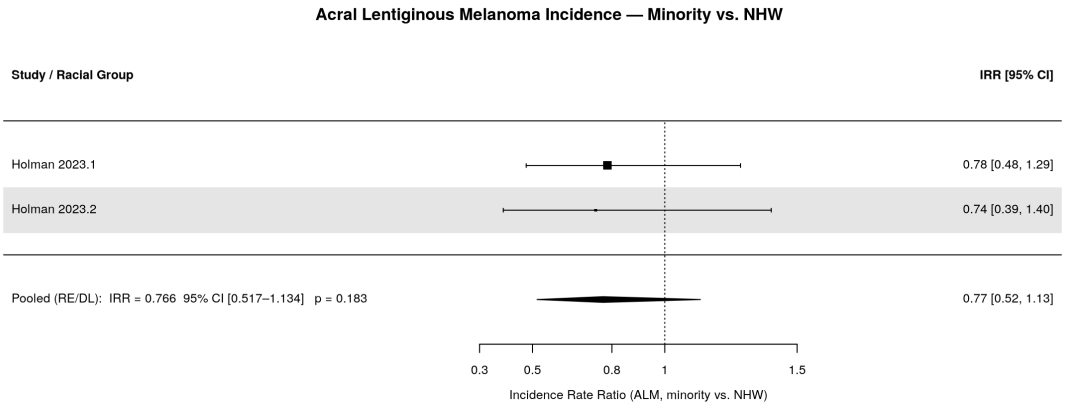

Figure 5. Melanoma Survival — Black vs. Non-Hispanic White (Hazard Ratio)

Adjusted HR for overall survival (Lam 2022: 1.42, 95% CI 1.25–1.60), melanoma-specific survival (Lam 2022: 1.27, 95% CI 1.03–1.56), and all-cause death (Zell 2008: 1.60, 95% CI 1.17–2.18). HR > 1.0 = worse survival in Black patients despite lower absolute incidence.

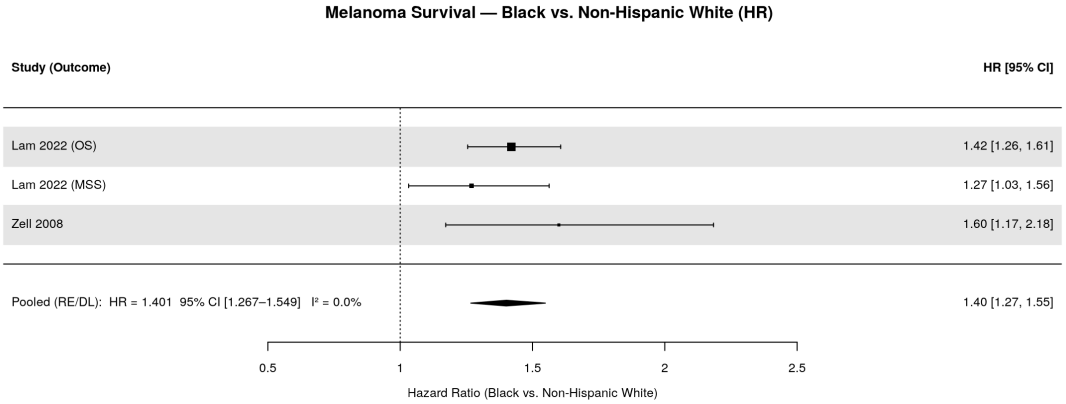

Figure 6. Summary Dot-and-Whisker Plot — Pooled IRRs by Race/Ethnicity

Pooled incidence rate ratios with 95% confidence intervals on a log scale. All comparisons significant at $p < 0.001$ except ALM ($p = 0.183$).

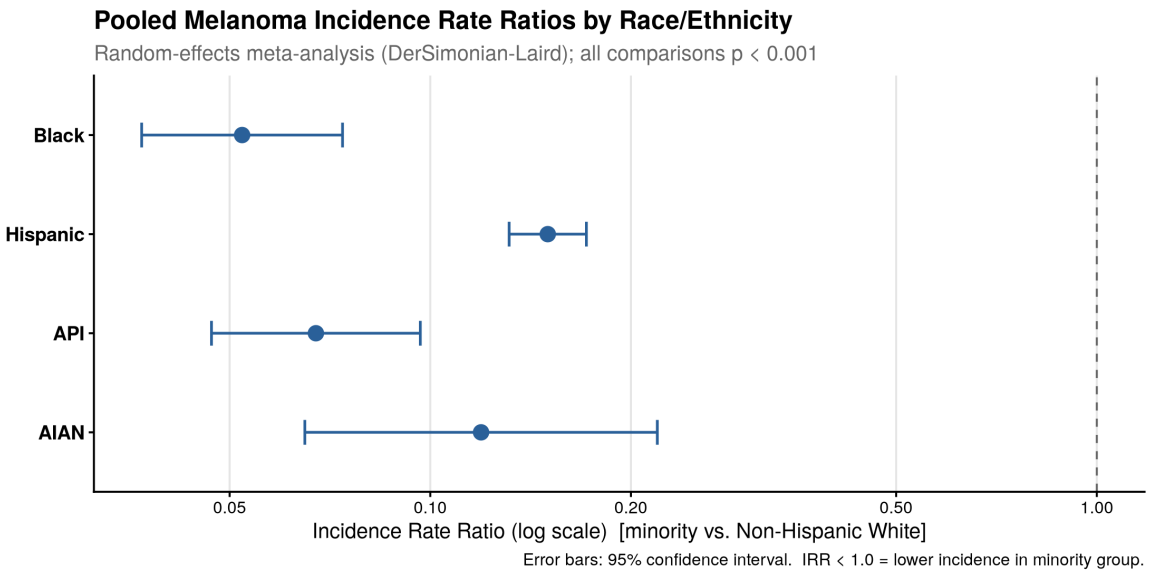

Figure 7. Funnel Plot — Black/African American vs. NHW (Melanoma Incidence)

Four contributing studies plotted as log(IRR) against standard error. No overt asymmetry; formal Egger's test was not performed due to k < 10.

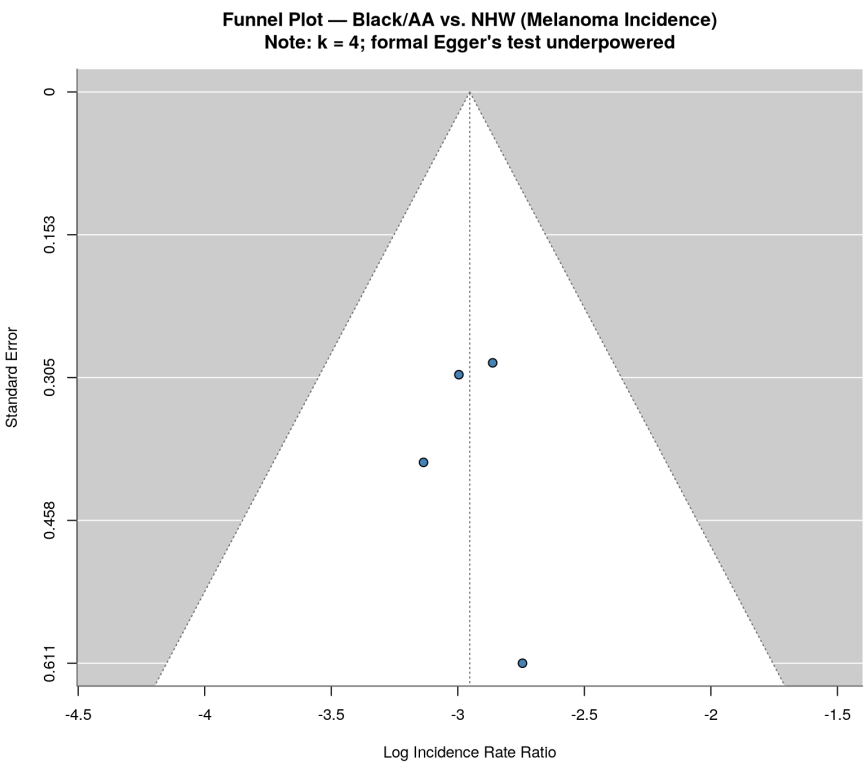

Figure Legends

Figure 1. PRISMA 2020 Flow Diagram.
Records identified from Embase (n = 107); records after duplicate removal (n = 43); records excluded at title/abstract screening (n = 15; reasons: non-skin cancer n = 6, treatment-only outcomes n = 4, review without primary data n = 3, case series n = 2); full-text articles assessed (n = 28); full-text articles excluded (n = 14; reasons: insufficient racial/ethnic stratification n = 6, outcome data not extractable n = 5, conference abstract without full publication n = 3); studies included in qualitative synthesis (n = 14); studies included in quantitative synthesis/meta-analysis (n = 13).

Figure 2. Risk of Bias Summary (Newcastle-Ottawa Scale).
Traffic light plot showing risk of bias ratings for the 14 included studies across three NOS domains: (1) selection of study groups; (2) comparability; (3) exposure/outcome ascertainment. Green = low risk; yellow = moderate risk. No studies rated high or critical risk.

Figure 3. Forest Plot: Melanoma Incidence Rate Ratios (Minority vs. Non-Hispanic White).
Random-effects meta-analysis (DerSimonian-Laird) for melanoma incidence IRRs comparing (A) Black/African American, (B) Hispanic/Latino, (C) Asian/Pacific Islander, and (D) American Indian/Alaska Native populations to Non-Hispanic White reference. Squares proportional to study weight; horizontal bars represent 95% CIs; diamond represents pooled estimate with 95% CI. $I^2 = 0\%$ for all comparisons.

Figure 4. Subgroup Analysis: Melanoma Incidence vs. ALM Incidence.
Forest plot comparing IRRs for overall melanoma incidence (k = 2–4 studies per group) with ALM-specific incidence (k = 2, Holman 2023) across minority groups. Demonstrates attenuation of IRR toward null for ALM compared with overall melanoma, consistent with UV-independent pathogenesis.

Figure 5. Funnel Plot — Melanoma Incidence, Black vs. NHW.
Funnel plot of log(IRR) against standard error for the Black vs. NHW melanoma incidence comparison (k = 4). Visual inspection reveals no overt asymmetry; formal statistical testing was not performed due to insufficient study numbers (k < 10).

Supplementary Materials

Supplementary Appendix S1. Full Embase search strategy (Ovid syntax), including all Emtree terms, Boolean operators, field tags, and date filters.

Supplementary Appendix S2. Adapted search strategies for PubMed, CINAHL, Cochrane Library, and Web of Science.

Supplementary Appendix S3. Newcastle-Ottawa Scale adapted for registry-based and ecological studies: pre-specified domain modifications.

Supplementary Table S1. Domain-level risk of bias assessments for all 14 included studies using the Newcastle-Ottawa Scale.

Supplementary Table S2. Complete GRADE evidence profile with detailed domain ratings and rationale for each outcome-comparison.

Word count (main text, excluding abstract, tables, references, figure legends): ~4,800 words

Generated by Meta-Analysis Agent System — Racial/Ethnic Disparities in Skin Cancer